

A Small Portable IoT HEALTH STICKER

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Monitoring health of a person continuously 24 hours a day and every day of the week is not easy for even doctors and researchers, unless the person is under care in a hospital or clinic. For a common person wishing to monitor his or her health in normal day to day life, it becomes even more difficult because most cardiac and other health monitors are not

portable. These cannot be left attached to a person's body whole day for continuous monitoring.

So, here is how you can design and make a small IoT-enabled monitor that you can stick to your body to capture your health data. It will capture data like heartbeat and body temperature in real time and show it on your phone or any other device that can be connected to Wi-Fi. It

can be used anywhere to monitor the health parameters on a continuous basis. The author's prototype is shown in Fig. 1.

The circuit and working

Fig. 2 shows circuit diagram of the IoT health sticker, which is built around ESP12F (MOD1), MAX 301002 (MOD2), and MCP73812T (MOD3) ICs. The circuit has two



Fig. 1: Author's prototype

BILL OF MATERIAL

Components	Quantity	Description
ESP12F (MOD1)	1	Wi-Fi MCU SMD chip
MAX 301002 (MOD2)	1	Health sensor module
MCP73812T (MOD3)	1	Cell charging SMD chip
LED1, LED2, LED3	3	SMD LED
4.7 μ F capacitor (C1, C2)	2	SMD capacitor
Resistor 100 Ω (R1)	1	SMD resistor
Resistor 470 Ω (R2)	1	SMD resistor
Resistor 2k Ω (R3)	1	SMD resistor
Slide switch (S1)	1	SPST slide switch
Jumper 2-pin male	1	Connector
3.3V battery	1	LiPo rechargeable

Note: In addition, you would need the USB FTDI or any programming board to upload the code to ESP12F chip.

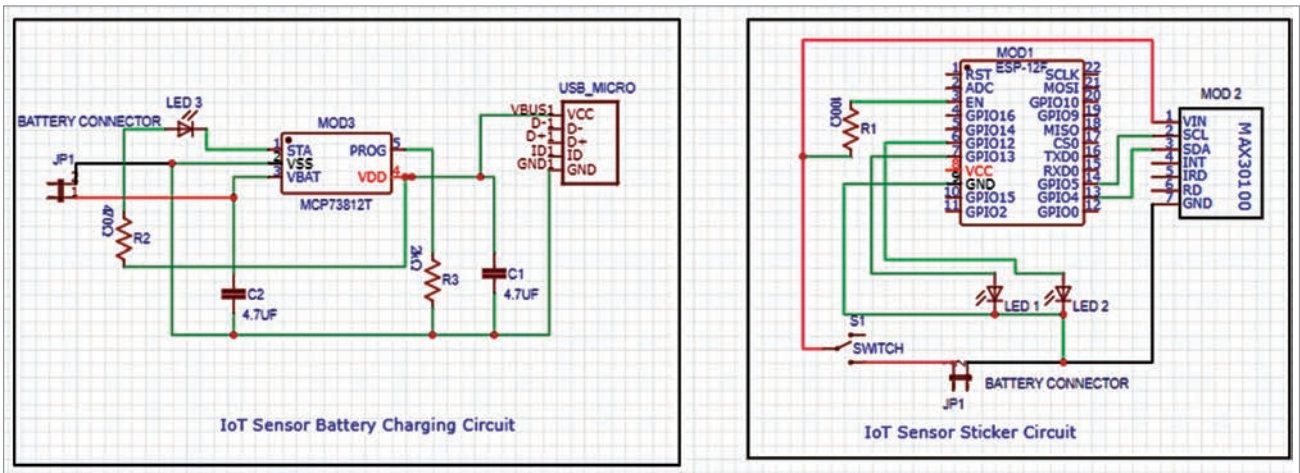


Fig. 2: The two parts of health sticker circuit